

Multivariate Analysis – Honours

Section 1 Exercises Solutions

Chapter 2

Homework exercise 2.1

a)

$$\mathbf{A}'\mathbf{A} = \begin{bmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{bmatrix} \begin{bmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{bmatrix} = \begin{bmatrix} 25 & 50 & 5 \\ 50 & 100 & 10 \\ 5 & 10 & 145 \end{bmatrix}$$

$$\begin{aligned} |\mathbf{A}'\mathbf{A} - \lambda\mathbf{I}| &= \begin{vmatrix} 25 - \lambda & 50 & 5 \\ 50 & 100 - \lambda & 10 \\ 5 & 10 & 145 - \lambda \end{vmatrix} \\ &= (25 - \lambda) \begin{vmatrix} 100 - \lambda & 10 \\ 10 & 145 - \lambda \end{vmatrix} - 50 \begin{vmatrix} 50 & 10 \\ 5 & 145 - \lambda \end{vmatrix} + 5 \begin{vmatrix} 50 & 100 - \lambda \\ 5 & 10 \end{vmatrix} \\ &= (25 - \lambda) [(100 - \lambda)(145 - \lambda) - 100] - 50 [50(145 - \lambda) - 50] + 5 [500 - 5(100 - \lambda)] \\ &= (25 - \lambda)(14400 - 245\lambda + \lambda^2) - 50(7200 - 50\lambda) + 25\lambda \\ &= -\lambda^3 + 270\lambda^2 - 18000\lambda \\ &= -\lambda(\lambda - 150)(\lambda - 120) \end{aligned}$$

The eigenvalues are $\lambda_1 = 150$, $\lambda_2 = 120$ and $\lambda_3 = 0$

For $\lambda_1 = 150$:

$$\mathbf{A}'\mathbf{A}\mathbf{x} = \lambda\mathbf{x} = 150\mathbf{x} = \begin{bmatrix} 25x_1 + 50x_2 + 5x_3 \\ 50x_1 + 100x_2 + 10x_3 \\ 5x_1 + 10x_2 + 145x_3 \end{bmatrix} = \begin{bmatrix} 150x_1 \\ 150x_2 \\ 150x_3 \end{bmatrix}$$

From the 3rd equation we get $x_3 = x_1 + 2x_2$, with the other two then yielding $x_2 = 2x_1$.

$$\therefore \begin{bmatrix} t \\ 2t \\ 5t \end{bmatrix} \text{ is an eigenvector for } \lambda_1 = 150 \implies \mathbf{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ \frac{5}{\sqrt{30}} \end{bmatrix}$$

For $\lambda_2 = 120$:

$$\mathbf{A}'\mathbf{A}\mathbf{x} = \lambda\mathbf{x} = 120\mathbf{x} = \begin{bmatrix} 25x_1 + 50x_2 + 5x_3 \\ 50x_1 + 100x_2 + 10x_3 \\ 5x_1 + 10x_2 + 145x_3 \end{bmatrix} = \begin{bmatrix} 120x_1 \\ 120x_2 \\ 120x_3 \end{bmatrix}$$

From the 3rd equation we get $x_1 = -2x_2 - 5x_3$, with the other two then yielding $x_2 = -2x_3$.

$$\therefore \begin{bmatrix} t \\ 2t \\ -t \end{bmatrix} \text{ is an eigenvector for } \lambda_2 = 120 \implies \mathbf{v}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix}$$

We will not require $\mathbf{v}_3$ when writing the SVD in compact form.

b)

$$\mathbf{A}\mathbf{A}' = \begin{bmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{bmatrix} = \begin{bmatrix} 144 & -12 \\ -12 & 126 \end{bmatrix}$$

$$\begin{aligned} |\mathbf{A}\mathbf{A}' - \lambda \mathbf{I}| &= \begin{vmatrix} 144 - \lambda & -12 \\ -12 & 126 - \lambda \end{vmatrix} \\ &= (144 - \lambda)(126 - \lambda) - 144 \\ &= \lambda^2 - 270\lambda + 18000 \\ &= (\lambda - 150)(\lambda - 120) \end{aligned}$$

The eigenvalues are $\lambda_1 = 150$ and $\lambda_2 = 120$, same as in part a).

For $\lambda_1 = 150$:

$$\mathbf{A}\mathbf{A}'\mathbf{x} = \lambda\mathbf{x} = 150\mathbf{x} = \begin{bmatrix} 144x_1 - 12x_2 \\ -12x_1 + 126x_2 \end{bmatrix} = \begin{bmatrix} 150x_1 \\ 150x_2 \end{bmatrix}$$

Yielding $x_1 = -2x_2$.

$$\therefore \begin{bmatrix} 2t \\ -t \end{bmatrix} \text{ is an eigenvector for } \lambda_1 = 150 \implies \mathbf{u}_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}$$

For $\lambda_2 = 120$:

$$\mathbf{A}\mathbf{A}'\mathbf{x} = \lambda\mathbf{x} = 120\mathbf{x} = \begin{bmatrix} 144x_1 - 12x_2 \\ -12x_1 + 126x_2 \end{bmatrix} = \begin{bmatrix} 120x_1 \\ 120x_2 \end{bmatrix}$$

Yielding $x_2 = 2x_1$.

$$\therefore \begin{bmatrix} t \\ 2t \end{bmatrix} \text{ is an eigenvector for } \lambda_2 = 120 \implies \mathbf{u}_2 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

c)

$$\begin{aligned} \mathbf{A} &= \mathbf{U}\mathbf{\Lambda}\mathbf{V}' \\ \begin{bmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{bmatrix} &= \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{150} & 0 \\ 0 & \sqrt{120} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{6}} & \frac{5}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \sqrt{150} \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{5}{\sqrt{30}} \end{bmatrix} + \sqrt{120} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \end{aligned}$$
